

AMRUTH S SHARMA

amruths604@gmail.com | 8050495260 | [linkedin.com/in/amruthssharma](https://www.linkedin.com/in/amruthssharma) | github.com/amruthssss | amruthssharmaportfolio.netlify.app

EDUCATION

Bachelor of Engineering in Computer Science and Engineering

Sai Vidya Institute of Technology, Bengaluru

2022 - 2026

CGPA: 8.2 / 10.0

Relevant Coursework: DBMS, Operating Systems, OOPS, Machine Learning, Artificial Intelligence, Data Analytics

TECHNICAL SKILLS

Languages: Python, SQL, C, HTML, CSS

Frameworks & Libraries: FastAPI, Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, Capacitor.js, LangChain

Machine Learning & AI: Generative AI, LLMs, RAG, Prompt Engineering, YOLOv11, Prophet, Agentic Systems, LSTM

Databases: MySQL, PostgreSQL, SQLite, MongoDB, Firestore, ChromaDB

Cloud & Deployment: Firebase, AWS (EC2, S3), Google Cloud Platform, Vercel, Streamlit Cloud

Tools & Platforms: Git, GitHub, Android Studio, OpenCV, Power BI, WebSockets, REST APIs, Google AI Studio

EXPERIENCE

MindMatrix.io | Android Developer Intern | Bengaluru, India

Feb 2026 – May 2026

- Developed Akshara-Deepa, a full stack Android SSLC learning app using Capacitor.js, Firebase, and Google Sign-In.
- Integrated Gemini 1.5 Flash for adaptive quizzes, answer evaluation, personalized study planner, and AI tutor.
- Built Strength Map radar charts visualizing subject mastery, accuracy, and streaks synced via Firestore.
- Owned the full development lifecycle — UI/UX design (Material Design), Android Studio setup, APK testing and debugging, and production deployment via Gradle; ensured a 100% crash-free launch on the Google Play internal test track.

PROJECTS

LaunchPad AI

FEB 2026

An autonomous multi-agent platform transforming a startup idea into market research, architecture, production-ready code, and ML-driven revenue forecasts. Built on FastAPI + React with WebSocket streaming, PostgreSQL persistence, and a self-correcting Critic Agent (LLM scoring 0.0–1.0) with LangGraph retry routing, grounded by a RAG pipeline (ChromaDB + OpenAI embeddings). Forecasting engine combines Facebook Prophet, XGBoost, and LSTM for zero-historical-data projections.

Skills: Python, LangGraph, LangChain, FastAPI, React, PostgreSQL, ChromaDB, XGBoost, LSTM, RAG, OpenAI GPT-4

GitHub: <https://github.com/amruthssss/LaunchPad-AI>

Multimodal Surveillance System for Intelligent Security Monitoring

JUN 2025

A real-time AI surveillance platform detecting fire, intrusion, explosion, accidents, and smoke from live feeds using parallel deep learning. Fused YOLOv11, 3D-CNN + LSTM action recognition, and CNN emotion recognition via an Adaptive Fusion Engine with Temporal Validation (65% confidence across 5 frames). Threat detection triggers an audio alarm and instant SMS/email notifications via Twilio. Events logged in MongoDB and streamed live via Flask + React dashboard.

Skills: Python, YOLOv11, 3D-CNN, LSTM, PyTorch, Flask, React, OpenCV, PostgreSQL

GitHub: <https://github.com/amruthssss/multimodal-Surveillance->

Orbit AI Engineering Platform

AUG 2026

A full-stack AI engineering platform integrating Gemini-powered chat, RAG, AI agents, tool calling, workflows, and LLM evaluation. Built with FastAPI + React, PostgreSQL, and Redis, with a RAG pipeline supporting document ingestion, Gemini embeddings, vector retrieval, reranking, and grounded responses with citations. Integrated Tavily for web research and deployed the frontend on Vercel and backend on Render with Supabase PostgreSQL and Upstash Redis.

Skills: Python, FastAPI, React, TypeScript, Gemini, LLMs, RAG, Embeddings, Vector Search, Reranking, AI Agents, Tool Calling, LLM Evaluation, PostgreSQL, Redis, Tavily

GitHub: https://github.com/amruthssss/Orbit_AI

EV Charging Analytics Platform

MAR 2025

An interactive ML platform tracking EV infrastructure usage, forecasting grid demand, and clustering patterns across 457 charging stations. Built a multi-page Streamlit app with Plotly dashboards and a predictive pipeline combining Facebook Prophet with Histogram Gradient Boosting and Random Forest regressors — isolating 4 station behavioural personas via K-Means and PCA.

Skills: Python, Streamlit, Facebook Prophet, Scikit-learn, K-Means, PCA, Plotly

GitHub: https://github.com/amruthssss/ev_charging_analysis

Retail Inventory and Sales Analytics Platform

OCT 2024

A full-stack retail dashboard providing real-time inventory visibility, AI-driven demand forecasting, and automated restock alerts. Built 8+ interactive Streamlit dashboards covering inventory tracking, expiry monitoring, and sales funnels; integrated Facebook Prophet forecasting, an optimized MySQL query layer, SMTP restock alerts, and a PDF report generator with login-based auth and admin management.

Skills: Python, Streamlit, MySQL, Pandas, Facebook Prophet, SMTP

GitHub: <https://github.com/amruthssss/retail-project>

Spoken Keyword Spotting System

SEP 2024

A lightweight keyword spotting system built with a hybrid CNN-SVM pipeline using Librosa MFCC and mel-spectrogram features, trained on Google Speech Commands. INT8 quantization reduced model size by 75% and latency by 40%, achieving a 0.98 F1-score across 10 keyword classes on edge devices.

Skills: Python, TensorFlow, Scikit-learn, Librosa, NumPy, CNN, SVM, INT8 Quantization

GitHub: <https://github.com/amruthssss/keySpotting/tree/master/Spoken-Keyword-Spotting>

CERTIFICATIONS

- Google AI Professional Certificate —Google (Coursera) | 2026
- Data Analytics with Python —NPTEL, IIT Roorkee | 2026
- Artificial Intelligence Fundamentals —IBM SkillsBuild | 2025
- Networking and Cloud Computing —Microsoft (Coursera) | 2025
- Python Data Analysis —Rice University (Coursera) | 2025